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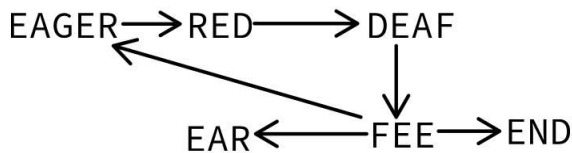
H. Word Chain

time limit per test: 1 second

memory limit per test: 256 megabytes

Bob is playing the word chain game alone. In this game, he has a list of n ($1 \leq n \leq 10^5$) words $w_1, \dots, w_n$ and starts with word A (which is also in the list).

If Bob's current word is X , he can move to another word Y from the list if the last letter of X is equal to the first letter of Y . For example, if the current word is **EAGER**, then the next word can be **RED** (if it is in the list) as the last letter of **EAGER** and the first letter of **RED** are both **R**.



Graph for the first sample.

Bob is wondering if he can reach the word B (which is also in the list).

Input

The first line contains the integer n and the words A and B (starting word and target word). The i -th line of the next n lines contains w_i , the i -th word.

All words consist of uppercase Latin letters only and are non-empty. It is guaranteed that the sum of length of all the words in the list is at most 10^6 .

Output

In a single line, output **YES** if it is possible to reach word B from word A . Otherwise, output **NO**.

Examples

input	Copy
6 EAGER EAR EAGER RED FEE DEAF END EAR	
output	Copy
YES	
input	Copy
5 EAGER EAR EAGER RED DEAF END EAR	

CSE221 Assignment 06 Summer 2026

Contest is running

3 days

Contestant


[→ Submit?](#)
Language: Java 21 64bit
 Choose file: Choose File No file chosen


Success!



Submit

output

Copy

NO

Note

For the first sample, Bob can reach **EAR** via the following path:

EAGER → **RED** → **DEAF** → **FEE** → **EAR**.

In the second sample, the word **FEE** has been removed. Now, it is impossible to reach **EAR** from **EAGER**.

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The only programming contests Web 2.0 platform

Server time: Aug/20/2026 17:18:29^{UTC+6} (i1).

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